

Waves and Sound — Complete Notes

Grade 11

Physical Sciences

CAPS

What you'll learn

- Describe a wave as a transfer of energy without transferring matter.
- Distinguish transverse from longitudinal waves.
- Define wavelength, frequency, period, amplitude and wave speed.
- Use the wave equation $v = f \lambda$ and the relation $f = 1 / T$.
- Explain how sound is produced, how it travels and how it is detected.
- Discuss pitch, loudness, echoes, ultrasound and resonance.

What is a wave?

A wave is a disturbance that transfers energy from one point to another without transferring the particles (matter) of the medium. The particles of the medium only vibrate about their rest positions; it is the energy that moves along.

Transverse and longitudinal waves

Feature	Transverse wave	Longitudinal wave
Particle motion	perpendicular to the direction of travel	parallel to the direction of travel
Examples	water ripples, a wave on a string	sound, a spring pushed and pulled
Features	crests and troughs	compressions and rarefactions

A transverse wave travelling to the right (particles move up and down):



A longitudinal wave travelling to the right (particles move left and right):



Wave quantities

Quantity	Symbol	Unit	Meaning
Wavelength	λ (lambda)	m	distance between two consecutive points in phase
Frequency	f	Hz (s^{-1})	number of complete waves per second
Period	T	s	time for one complete wave; $T = 1 / f$
Amplitude	A	m	maximum displacement from the rest position
Wave speed	v	$\text{m} \cdot \text{s}^{-1}$	distance the wave travels per second

Points are in phase when they are separated by one wavelength (or a whole number of wavelengths) and move in the same direction.

The wave equation

$$v = f \lambda$$

Because $f = 1 / T$, we can also write $v = \lambda / T$.

The speed of a wave depends on the medium, not on f or λ . When a wave moves into a new medium, its speed and wavelength change, but its frequency stays the same.

Sound

- Sound is a longitudinal wave produced by a vibrating source.
- It needs a medium (solid, liquid or gas) — it cannot travel through a vacuum.
- It travels fastest in solids, slower in liquids, and slowest in gases. In a solid the particles are packed closely together, so they pass the vibration on more quickly.
- In air at about 25 °C, the speed of sound is about $340 \text{ m} \cdot \text{s}^{-1}$.
- Compressions are regions of high pressure; rarefactions are regions of low pressure.

Sound properties:

- Pitch (how high or low a note sounds) depends on frequency. A higher frequency gives a higher pitch.
- Loudness depends on amplitude. A larger amplitude gives a louder sound.
- The speed of sound in air depends slightly on the temperature of the air.

Echoes, ultrasound and sonar

An echo is a sound wave reflected off a surface. If the total time for the sound to travel to the surface and back is t, then the distance to the surface is:

$$d = (v \times t) / 2$$

Ultrasound has a frequency above the human hearing range (above about 20 000 Hz). It is used in medicine, in cleaning and in sonar. Infrasound is below about 20 Hz, below the range of human hearing.

Resonance

Every object has a natural frequency at which it vibrates most easily. When a driving frequency matches an object's natural frequency, the amplitude of vibration becomes large — this is resonance. Example: a wine glass can shatter when a sound matches its natural frequency.

Worked example 1 — the wave equation

A wave on a string has a frequency of 250 Hz and a wavelength of 1,4 m. Calculate its speed.

Given: $f = 250 \text{ Hz}$; $\lambda = 1,4 \text{ m}$.

$$v = f \lambda = 250 \times 1,4 = 350 \text{ m}\cdot\text{s}^{-1}$$

Worked example 2 — speed of sound and an echo

A hiker claps her hands and hears the echo from a cliff 1,2 s later. If the speed of sound in air is $340 \text{ m}\cdot\text{s}^{-1}$, how far away is the cliff?

The sound travels to the cliff and back, so the total distance is $2d$.

$$\text{total distance} = v \times t = 340 \times 1,2 = 408 \text{ m}$$

$$d = 408 / 2 = 204 \text{ m}$$

Worked example 3 — period and frequency

A sound wave has a period of $2,5 \times 10^{-3} \text{ s}$.

(a) Calculate its frequency. (b) If the speed of sound is $340 \text{ m}\cdot\text{s}^{-1}$, calculate its wavelength.

$$(a) f = 1 / T = 1 / (2,5 \times 10^{-3}) = 400 \text{ Hz}$$

$$(b) \lambda = v / f = 340 / 400 = 0,85 \text{ m}$$

Speed of sound in different media

Medium	Speed of sound ($\text{m}\cdot\text{s}^{-1}$)
Air (25 °C)	≈ 340
Water	$\approx 1\,500$
Steel	$\approx 5\,000$

Sound travels faster when the particles of the medium are held more tightly together. This is why you hear an approaching train through the rails before you hear it through the air.

Ultrasound in medicine and industry

Because ultrasound waves have short wavelengths, they are reflected by small objects and can be used to build up pictures. Medical scans use ultrasound to view organs and unborn babies, and ultrasound is also used to clean delicate instruments and to detect cracks inside metal. Bats and dolphins use ultrasound echoes to locate food and obstacles.

Worked example 4 — hearing thunder

A learner sees a flash of lightning and hears the thunder 5,0 s later. If the speed of sound is $340 \text{ m}\cdot\text{s}^{-1}$, how far away did the lightning strike?

$$\text{distance} = v \times t = 340 \times 5,0 = 1\,700 \text{ m}$$

The light from the flash reaches the learner almost instantly, so the 5,0 s is the time the sound takes to travel the whole distance.

Common mistakes

- Using $v = f \lambda$ with λ in cm and v in $\text{m}\cdot\text{s}^{-1}$ without converting ($1 \text{ m} = 100 \text{ cm}$).
- Thinking sound can travel through a vacuum — it cannot; it always needs a medium.
- Forgetting that an echo travels to the surface and back, so the distance must be halved.
- Believing a higher amplitude means a higher pitch. Amplitude controls loudness; frequency controls pitch.
- Confusing frequency (Hz) with period (s). Remember $f = 1 / T$.
- Saying a wave transfers particles. It transfers energy, while the particles vibrate in place.
- Assuming the speed changes when the frequency changes. In the same medium, the speed stays constant.

Quick summary / cheat sheet

- Waves transfer energy, not matter.
- Transverse: particles move perpendicular to travel. Longitudinal: particles move parallel (sound).
- $v = f \lambda$; $f = 1 / T$; λ in m, f in Hz, v in $\text{m}\cdot\text{s}^{-1}$.
- Speed of sound in air $\approx 340 \text{ m}\cdot\text{s}^{-1}$; sound needs a medium.
- Echo distance: $d = v t / 2$.
- Pitch $\leftrightarrow$ frequency; loudness $\leftrightarrow$ amplitude.
- Human hearing range $\approx 20 \text{ Hz}$ to $20\,000 \text{ Hz}$; above it is ultrasound, below it is infrasound.

Practice questions

1. Define a transverse wave and give one example. [2]
2. A water wave has a wavelength of 2,0 m and a frequency of 0,50 Hz. Calculate its speed. [2]
3. A sound wave travels through steel at $5\,000 \text{ m}\cdot\text{s}^{-1}$ with a frequency of 2 000 Hz. Calculate its wavelength. [2]
4. Explain why sound travels faster in a solid than in a gas. [2]
5. A ship sends a sonar pulse downwards. It returns after 0,80 s. If the speed of sound in seawater is $1\,500 \text{ m}\cdot\text{s}^{-1}$, calculate the depth of the water below the ship. [3]
6. A tuning fork vibrates with a period of $4,0 \times 10^{-3} \text{ s}$. Calculate its frequency. [2]
7. State the relationship between the loudness and the amplitude of a sound wave. [2]

Answers

1. A transverse wave is a wave in which the particles of the medium vibrate perpendicular to the direction of wave travel. Example: a wave on a string (or a water ripple).
1. $v = f \lambda = 0,50 \times 2,0 = 1,0 \text{ m}\cdot\text{s}^{-1}$.
1. $\lambda = v / f = 5\,000 / 2\,000 = 2,5 \text{ m}$.

1. In a solid the particles are packed closely together, so they pass on the vibration (energy) to neighbouring particles more quickly than in a gas, where the particles are far apart.
1. total distance = $v t = 1\,500 \times 0,80 = 1\,200$ m; depth = $1\,200 / 2 = 600$ m.
1. $f = 1 / T = 1 / (4,0 \times 10^{-3}) = 250$ Hz.
1. Loudness increases as amplitude increases — loudness is directly related to the amplitude of the wave.

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